Status	Finished
Started	Sunday, 18 October 2026, 7:29 AM
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Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It only works with integer arrays
b. It requires external libraries
c. It sorts data out of the box efficiently
d. It is less efficient than custom sorting algorithms

The correct answer is: It sorts data out of the box efficiently

Question **2**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

a. Searching
b. Rearranging
c. Sorting
d. Merging

The correct answer is: Sorting

Question **3**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

- a. Merge sort
- b. Heap sort
- c. Quick sort
- d. Selection sort

The correct answer is:
Merge sort

Question **4**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 200,470,80,150,40,90,300,400,70,120
- b. 40,70,80,90,120,150,200,300,400,470
- c. 40,80,90,150,200,300,400,470,70,120
- d. 80,150,200,470,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **5**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

- a. Bubbling up the smallest element
- b. Sorting the entire list in one pass
- c. Dividing the list into halves
- d. Bubbling up the largest element to its correct position

The correct answer is: Bubbling up the largest element to its correct position

Question **6**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Linear Search
b. Bubble Sort
c. Merge Sort
d. Binary Search

The correct answer is: Bubble Sort

Question **7**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Merge Sort
b. Heap Sort
c. Quick Sort
d. Bubble Sort

The correct answer is: Bubble Sort

Question **8**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Decreases the efficiency of algorithms
b. Provides a basis for other algorithms to work efficiently
c. Makes code execution slower
d. Makes data harder to manage

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **9**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

- a. It allows for quicker identification of duplicates
- b. It has no effect on the analysis process
- c. It complicates the analysis process
- d. It makes the analysis slower

The correct answer is: It allows for quicker identification of duplicates

Question **10**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

- a. It slows down the process
- b. It makes it easier to select items based on their relationship to the rest of the items
- c. It makes the data unsorted
- d. It complicates the selection of items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **11**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

- a. By breaking the input into smaller parts and merging them
- b. By sorting data in a single pass
- c. By comparing elements sequentially
- d. By using the bubble-up method

The correct answer is: By breaking the input into smaller parts and merging them

Question **12**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

- a. Divide and conquer
- b. Greedy method
- c. Dynamic programming
- d. Backtracking

The correct answer is:
Divide and conquer

Question **13**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. sorted()
- b. sort()
- c. arrange()
- d. order()

The correct answer is: sorted()

Question **14**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

- a. Bubble Sort
- b. Binary Search
- c. Linear Search
- d. Merge Sort

The correct answer is: Merge Sort

Question **15**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It sorts data using a single pass
b. It is based on the divide-and-conquer approach
c. It does not require recursion
d. It is less efficient than bubble sort

The correct answer is: It is based on the divide-and-conquer approach